Towards Synthetic Generation of Daily Routines and Anomaly Detection with Wearable Sensors for Dementia Care

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INTRODUCTION

Dementia requires continuous and unobtrusive monitoring of daily behaviour to support early detection of functional changes. Wearable technologies enable real-world acquisition of physiological and behavioural data for this purpose [1].
As shown in Figure 1, the proposed system integrates a wrist-worn [2] and an in-ear [3] device to collect multimodal data, which are processed by an AI engine for behavioural modelling and anomaly detection, supporting remote monitoring by caregivers in home and residential care contexts.

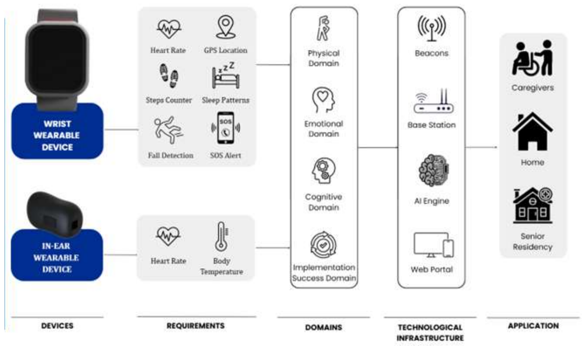


Fig. 1 - Overview of the wearable-based monitoring system for dementia care.

METHODOLOGY

1. SYNTHETIC ROUTINE GENERATION

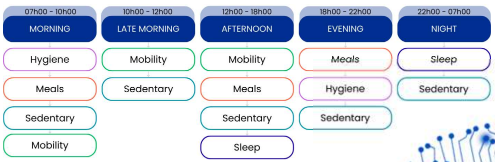
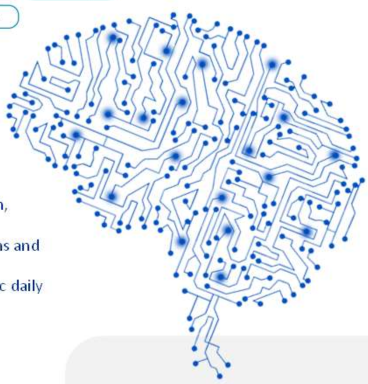


Fig. 2 - Time-of-day segmentation and typical distribution of core activity types used for synthetic routine generation.

- Each day is divided into 280 time steps (5-minute intervals).
- Five time blocks: morning, late morning, afternoon, evening, night, Figure 2.
- Activities generated using probabilistic distributions and typical duration ranges.
- Temporal variability introduced to simulate realistic daily behaviour.



2. ANOMALY SIMULATION

- 15% of daily steps are replaced by context-inappropriate activities.
- Examples include mobility at night, missing meals, and unexpected sleep.
- Each time step is labelled as normal (0) or anomalous (1), Figure 3.

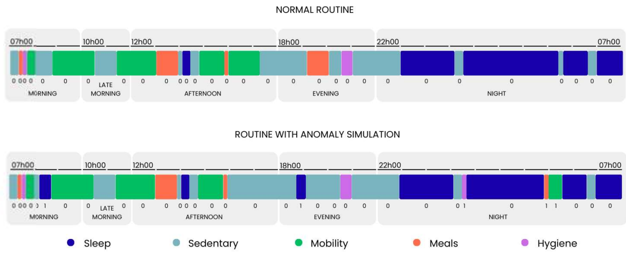


Fig. 3 - Time-of-day segmentation and typical distribution of core activity types used for synthetic routine generation.

3. ANOMALY DETECTION MODEL

- Bidirectional GRU (BiGRU) with activity embeddings, Figure 4.
- Input: daily activity sequences.
- Output: anomaly probability per time step.

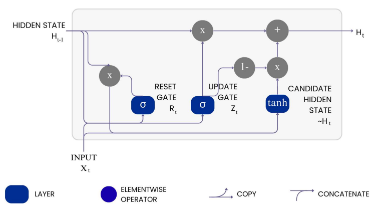


Fig. 4 - Time-of-day segmentation and typical distribution of core activity types used for synthetic routine generation.

RESULTS

As shown in Figure 5, the proposed model achieved high step-level classification performance, with an F1-score of 0.959 for anomalous activities and 0.997 for normal activities, indicating robust discrimination between normal and abnormal behaviour.

	Precision	Recall	F1-score
Normal	0,995	0,998	0,997
Anomalous	0,978	0,941	0,959

Fig. 5 - Performance metrics for normal and anomalous classes

The confusion matrix (Figure 6) shows that 92.50% of normal steps and 7.05% of anomalous steps were correctly classified, with very low misclassification rates, confirming the model's reliability under class imbalance.

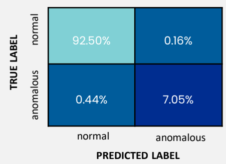


Fig. 6 – Confusion matrix (in %) summarizing step-level classification of normal and anomalous activities.

CONCLUSION AND FUTURE WORK

This work demonstrates that synthetic daily routine generation, combined with GRU-based anomaly detection, is an effective strategy for modelling behavioural patterns and identifying context-sensitive deviations in dementia care.
Future work will focus on:

- validating the approach using real-world wearable sensor data, and
- adapting the model for real-time, unidirectional inference in continuous monitoring systems.

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